

E Regression Analysis

Table 4: Regression Analysis for Agreement Sycophancy Based on Synthetic Interactions as Context

	Claude Sonnet 4	GPT 4.1 Mini	GPT 5.1	Gemini 2.5 Pro	Llama 4 Scout
context (β_1)	-0.006 (0.007)	0.053*** (0.009)	-0.088*** (0.009)	0.092*** (0.013)	0.152*** (0.018)
Intercept	0.005 (0.005)	0.809*** (0.055)	0.342*** (0.085)	0.082*** (0.029)	0.520*** (0.062)
N (Observations)	1,520	1,520	1,520	1,520	890
N (Participants)	38	38	38	38	17
R ²	0.749	0.452	0.531	0.511	0.360
Adjusted R ²	0.747	0.449	0.528	0.508	0.353

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 5: Regression Analysis for Agreement Sycophancy Based on User Interactions as Context

	Claude Sonnet 4	GPT 4.1 Mini	GPT 5.1	Gemini 2.5 Pro	Llama 4 Scout
context (β_1)	0.021 (0.027)	0.045** (0.020)	0.049 (0.035)	0.117*** (0.032)	0.247*** (0.033)
context x understanding (β_2)	-0.002 (0.011)	0.019* (0.006)	—	—	—
context x is_man (β_3)	0.030 (0.036)	-0.005 (0.023)	-0.036 (0.040)	-0.060* (0.035)	-0.054 (0.053)
context x is_liberal (β_4)	0.055* (0.029)	-0.025 (0.023)	-0.042 (0.045)	0.014 (0.038)	-0.010 (0.035)
context x is_man x is_liberal (β_5)	-0.055 (0.046)	0.006 (0.031)	0.046 (0.056)	0.041 (0.055)	0.077 (0.061)
Intercept	0.049* (0.025)	0.792*** (0.051)	0.630*** (0.040)	-0.002 (0.019)	0.532*** (0.057)
N (Observations)	1,520	1,520	1,520	1,520	1,070
N (Participants)	38	38	38	38	23
R ²	0.693	0.470	0.477	0.498	0.359
Adjusted R ²	0.690	0.465	0.472	0.494	0.351

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 6: Regression Analysis for Agreement Sycophancy Based on User Memory Profiles as Context

	Claude Sonnet 4	GPT 4.1 Mini	GPT 5.1	Gemini 2.5 Pro	Llama 4 Scout
context (β_1)	0.327*** (0.042)	0.164*** (0.013)	0.028 (0.024)	0.450*** (0.027)	-0.063* (0.036)
context x understanding (β_2)	-0.009 (0.016)	0.0001 (0.005)	—	—	—
context x is_man (β_3)	-0.021 (0.047)	-0.027 (0.018)	0.026 (0.034)	-0.076* (0.039)	-0.022 (0.043)
context x is_liberal (β_4)	0.045 (0.048)	-0.011 (0.014)	0.058 (0.048)	-0.029 (0.042)	0.023 (0.044)
context x is_man x is_liberal (β_5)	0.026 (0.084)	0.036 (0.027)	-0.050 (0.070)	0.132* (0.071)	0.055 (0.058)
Intercept	0.228*** (0.070)	0.781*** (0.046)	0.305*** (0.096)	0.257*** (0.066)	0.923*** (0.077)
N (Observations)	1,520	1,520	1,520	1,520	1,520
N (Participants)	38	38	38	38	38
R ²	0.406	0.477	0.422	0.470	0.392
Adjusted R ²	0.400	0.472	0.417	0.465	0.386

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ **Table 7: Regression Analysis for Perspective Sycophancy Based on User Interactions as Context**

	Political Explanations		Personal Advice	
	Claude Sonnet 4	GPT 4.1 Mini	Claude Sonnet 4	GPT 4.1 Mini
context (β_1)	0.179 (0.233)	-0.044 (0.156)	-0.118 (0.184)	-0.318 (0.216)
context x understanding (β_2)	0.204*** (0.078)	0.124** (0.058)	0.184*** (0.056)	-0.012 (0.068)
context x is_man (β_3)	-0.493 (0.302)	-0.169 (0.170)	-0.077 (0.205)	0.362 (0.226)
context x is_liberal (β_4)	-0.315 (0.241)	-0.031 (0.180)	-0.286 (0.180)	0.138 (0.250)
context x is_man x is_liberal (β_5)	0.472 (0.384)	-0.070 (0.266)	0.213 (0.255)	-0.392 (0.347)
is_man	0.253 (0.163)	0.144 (0.179)	-0.163 (0.144)	-0.074 (0.155)
is_liberal	0.140 (0.159)	0.037 (0.174)	-0.101 (0.146)	-0.342** (0.158)
Intercept	2.642*** (0.200)	3.217*** (0.217)	3.358*** (0.154)	3.285*** (0.175)
N (Observations)	380	380	380	380
N (Participants)	38	38	38	38
R ²	0.112	0.134	0.160	0.114
Adjusted R ²	0.073	0.095	0.123	0.075

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 8: Regression Analysis for User Rating of Understanding

	Political Explanations		Personal Advice	
	Claude Sonnet 4	GPT 4.1 Mini	Claude Sonnet 4	GPT 4.1 Mini
is_man	-0.010 (0.562)	-0.125 (0.582)	0.115 (0.400)	0.067 (0.226)
is_liberal	0.375 (0.579)	0.239 (0.582)	-0.227 (0.520)	-0.716 (0.513)
is_man x is_liberal	-0.990 (0.810)	-0.905 (0.804)	-0.055 (0.720)	-0.310 (0.756)
Intercept	3.625*** (0.492)	4.125*** (0.510)	4.500*** (0.374)	4.625*** (0.181)
N (Participants)	38	38	38	38
R ²	0.083	0.090	0.024	0.151
Adjusted R ²	0.002	0.010	-0.062	0.076

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ **Table 9: BH Correction for β_1 and Agreement Sycophancy**

Context	Model	β_1	SE	P-Value	BH Threshold	Significant?
Synthetic Interactions	GPT 4.1 Mini	0.053	0.009	<1e-4	0.0033	Yes
Synthetic Interactions	GPT 5.1	-0.088	0.009	<1e-4	0.0067	Yes
Synthetic Interactions	Gemini 2.5 Pro	0.092	0.013	<1e-4	0.0100	Yes
Synthetic Interactions	Llama 4 Scout	0.152	0.018	<1e-4	0.0133	Yes
User Interactions	Llama 4 Scout	0.247	0.033	<1e-4	0.0167	Yes
User Memory Profiles	GPT 4.1 Mini	0.164	0.013	<1e-4	0.0200	Yes
User Memory Profiles	Claude Sonnet 4	0.327	0.042	<1e-4	0.0233	Yes
User Memory Profiles	Gemini 2.5 Pro	0.450	0.027	<1e-4	0.0267	Yes
User Interactions	Gemini 2.5 Pro	0.117	0.032	0.0003	0.0300	Yes
User Interactions	GPT 4.1 Mini	0.045	0.020	0.0259	0.0333	Yes
User Memory Profiles	Llama 4 Scout	-0.063	0.036	0.0835	0.0367	No
User Interactions	GPT 5.1	0.049	0.035	0.1617	0.0400	No
User Memory Profiles	GPT 5.1	0.028	0.024	0.2420	0.0433	No
Synthetic Interactions	Claude Sonnet 4	-0.006	0.007	0.3872	0.0466	No
User Interactions	Claude Sonnet 4	0.021	0.027	0.4367	0.0500	No

Table 10: BH Correction for β_2 and Perspective Sycophancy

Task	Model	β_2	SE	P-Value	BH Threshold	Significant?
Personal Advice	Claude Sonnet 4	0.184	0.056	0.0011	0.0125	Yes
Political Explanations	Claude Sonnet 4	0.204	0.078	0.0088	0.025	Yes
Political Explanations	GPT 4.1 Mini	0.124	0.058	0.0331	0.0375	Yes
Personal Advice	GPT 4.1 Mini	-0.012	0.068	0.8548	0.05	No